Which Customers Come Back, Which Products Get Cancelled, and Where Free Shipping Matters

Portfolio Project 1 · SQL, SQLite, Excel · Online retail data

The problem

The three teams of an online store have three questions each. Marketing wants to know if it's worth testing a day 14 follow up email. Merchandising asks - which products are cancelled the most? The e-commerce team would like to understand the impact of a free shipping threshold. I had to perform the role of the analyst who is supposed to answer all three questions from one transaction file.

The data

I used the Kaggle Online Retail Dataset: 541,909 invoice lines from 1 December 2010 to 9 December 2011, in 8 columns (invoice number, product code, description, quantity, invoice date, unit price, customer ID and country). It is just one flat table, each row represents one product line within an invoice, it does not count the number of orders. The count of orders is the amount of unique invoice numbers.

Step 1: Asking questions the data can answer

I was wanting to see data on the file that it does not have: profit per customer, checkout conversion, whether or not customers "would have come back anyway". There is a file that has no cost, no session of a web site and no experiment. For every question, I changed the focus to a measurable and a clear decision:

1. Marketing - What percentage of first-time customers who didn't reorder by day 14 reordered from day 15 to day 60?
2. Merchandising - What are the items in the store that are most likely to be lost?
3. Ecommerce – how many orders are just one shy of each candidate free shipping level?

Step 2: Preparing and cleaning

I went through the data and did a profile before I did anything with it. Of the 137,965 lines, 25% (135,080) did not have a Customer ID, 9,288 lines were Cancellations and 11,805 lines contained a quantity of zero or less, or a price of zero or less. I did not make any changes to the raw table. On the other hand, I created a view to exclude the cancelled rows and the non-positive rows (which leaves 530,104) and another view with one row per invoice and the invoice value.

Step 3: Findings

Repeat purchases. Of 3,359 first-time customers who had not reordered by day 14, 27.7% (932) came back between day 15 and day 60. Another 10.2% of eligible customers had already reordered within 14 days.

Cancellations. Among products with at least 50 sales lines, the highest cancellation rates were the Savoy Art Deco Clock (24.5%), White Tea, Coffee, Sugar Jars (22.0%) and the Retro Plastic Elephant Tray (20.0%). Across the store, cancellation lines are roughly 2% of sales lines, so these products cancel at about ten times the average.

Free-shipping thresholds. In July to September 2011 there were 4,673 orders. The share of orders below a threshold rises quickly (13.2% below 100, 44.9% below 300), but the share just below it, within 20%, stays between 2.3% and 7.4% at every threshold I tested.

Three decisions that shaped the results

1. **Not a cause, a baseline! History of orders won't indicate if an e-mail would succeed. What it can demonstrate is how often they will return without one. The benchmark which a future test will need to outperform is 27.7%.**
2. **Lines, not units. The very first glance at the cancellations revealed single lines of 80995 and 74494 that appeared like huge orders entered and reversed. If it would have been a units-based rate, they would have dominated; therefore, I measured cancellation by line count and took out non-product codes.**
3. **Look before choosing a threshold.** I first assumed a threshold of 50, but only 7.7% of orders fall below it, so it would affect almost no one. Looking at the spread of order values first led me to test thresholds from 100 to 300.

Recommendations

1. Marketing: For each of the following, apply the day-14 email to a random group of people, then compare to the 27.7% baseline and compare discount cost. Not many customers are eligible, and only every month, so you need to consider the sample size first.

2. Merchandising: Start by reviewing the 3 most cancelled products, discuss these with Customer Service and begin to write down a reason for each cancellation.
3. Ecommerce: Only a few will be influenced by a threshold change. Get shipping-cost data and build a model for profit at 150 and 200, and test one level of the thresholds on a portion of customers.

How I used generative AI

i used an LLM that provided the analysis queries, and I did the concept explanations and ran all of the queries in an actual database. I didn't accept anything on faith. I compared row counts with the spreadsheet, I re-totalled totals by hand (382+1,166-234 = 1,314 repeat customers, and these matched the first query, for instance), I hand-checked individual customers/individual products, and I checked that order bands totaled the same as the total of the individual order counts (4,673). I also spotted two instances I myself had made: an unsuitable example threshold and a verification query with a place holder product code.

What I would do next

- Add shipping and product cost data so the threshold question can be answered in profit terms.
- Collect cancellation reasons so merchandising knows why, not just where.
- Design the email A/B test properly, including sample size.
- Build a Tableau dashboard of the three findings.

What I learned

• The first step in a good business question is to determine whether the data can be used to answer it.

• Counting rows is not the same as counting orders, customers or products. Understand the significance of one row at all times.

- Verifying involves checking against something other than the same tool, not asking it again.

Code, queries and the full write-up are in the GitHub repository sql-retail-business-analysis.